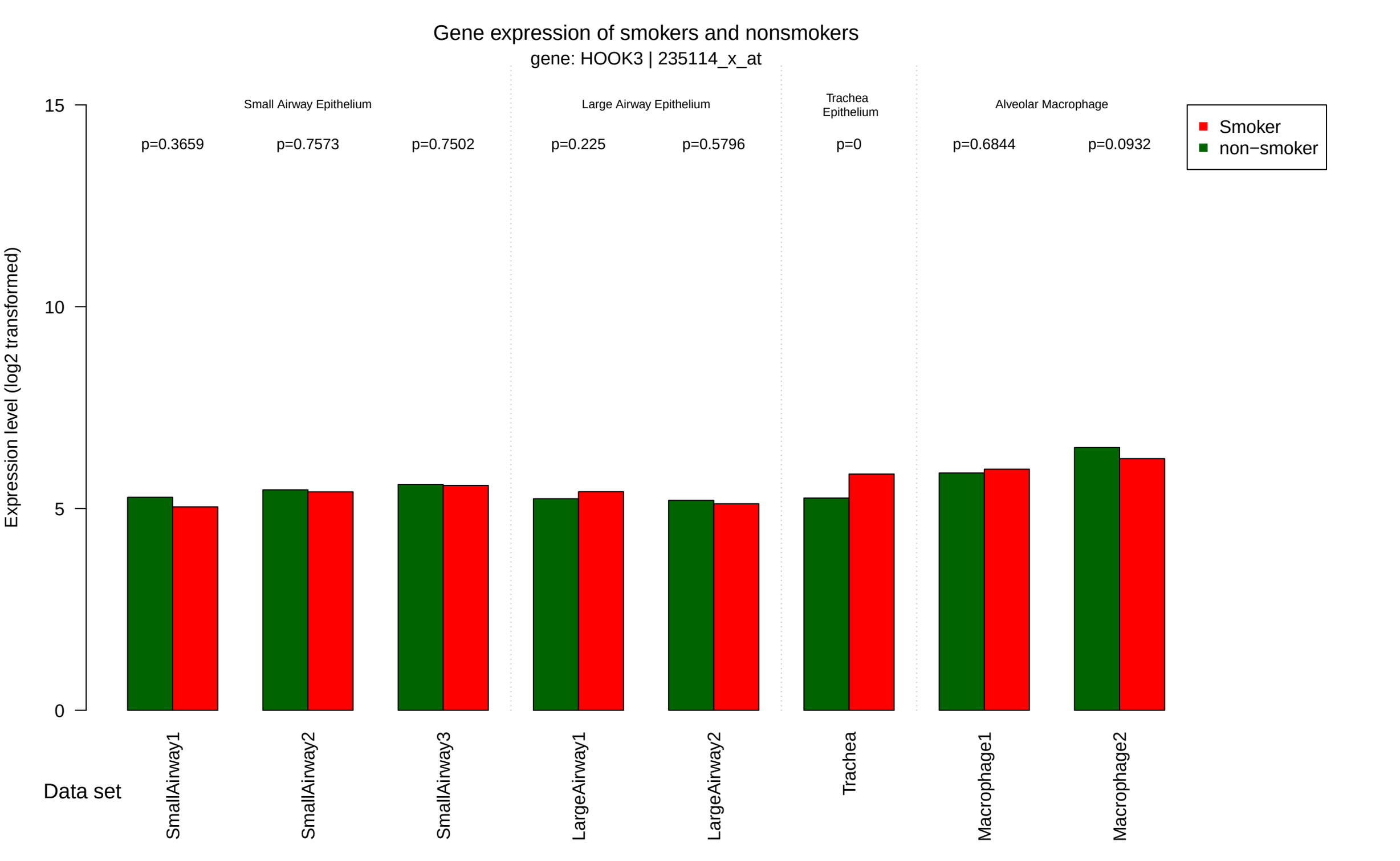
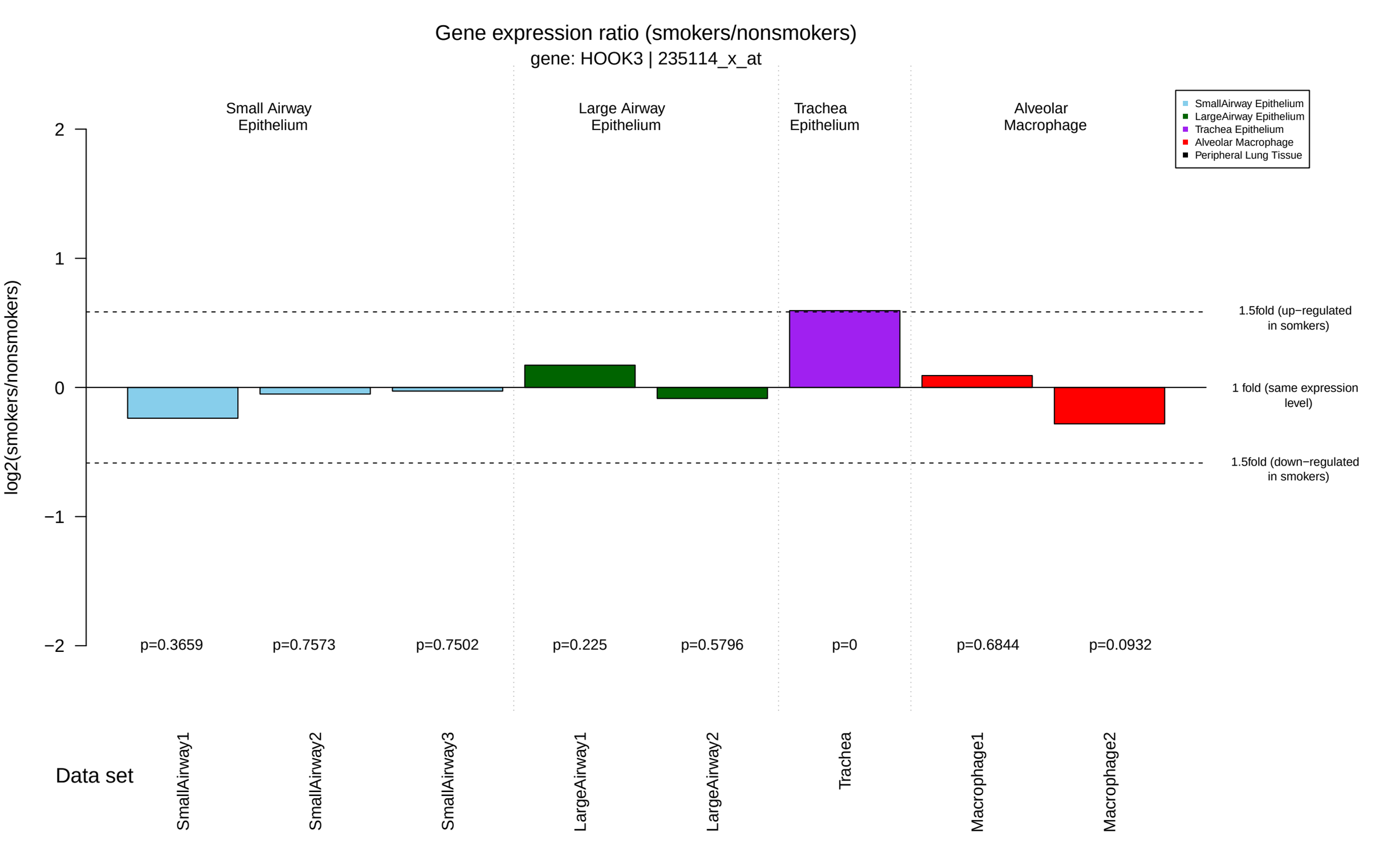
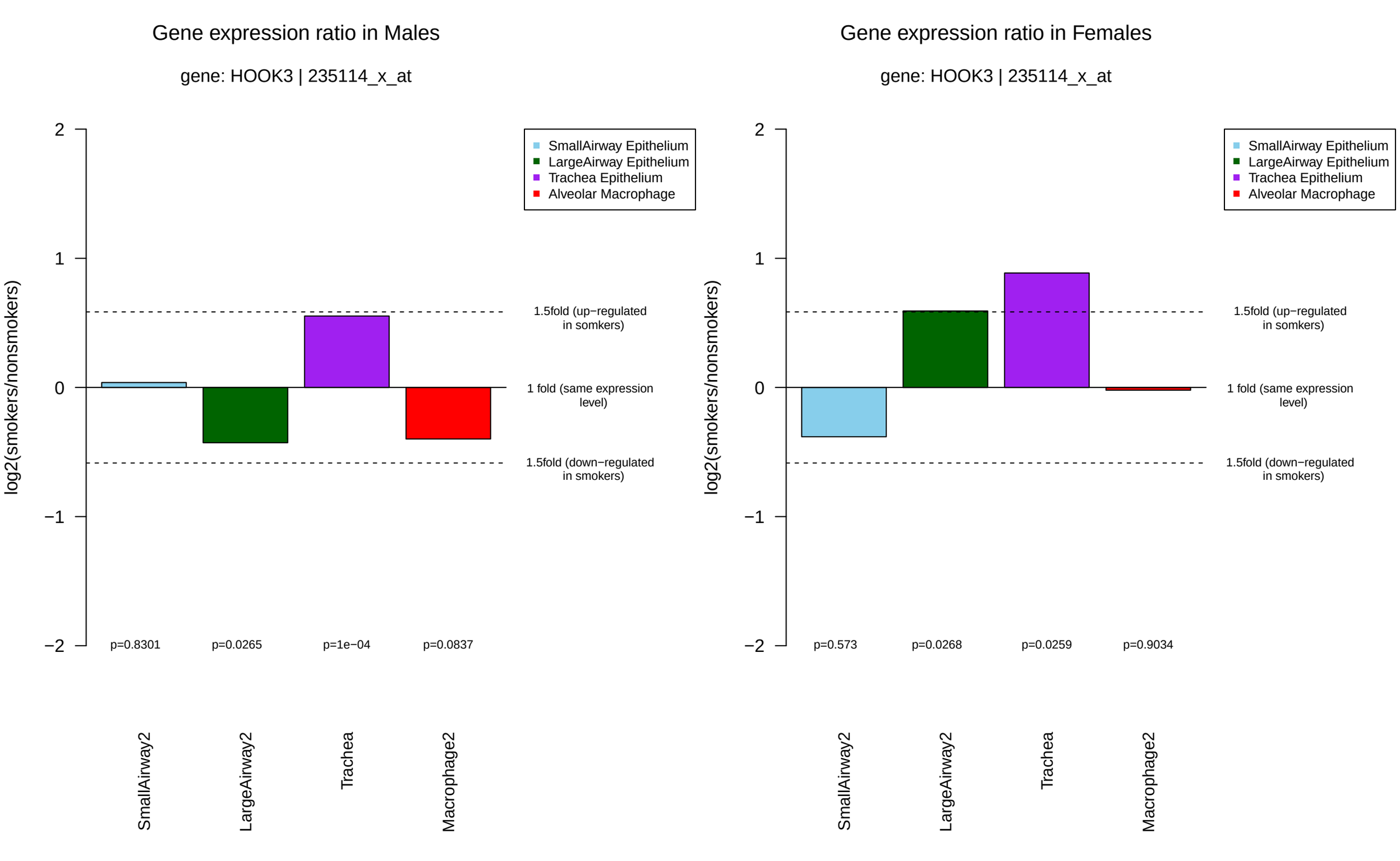
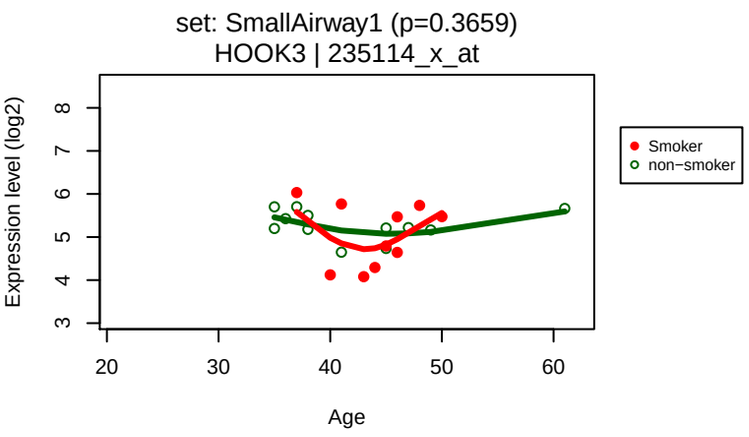
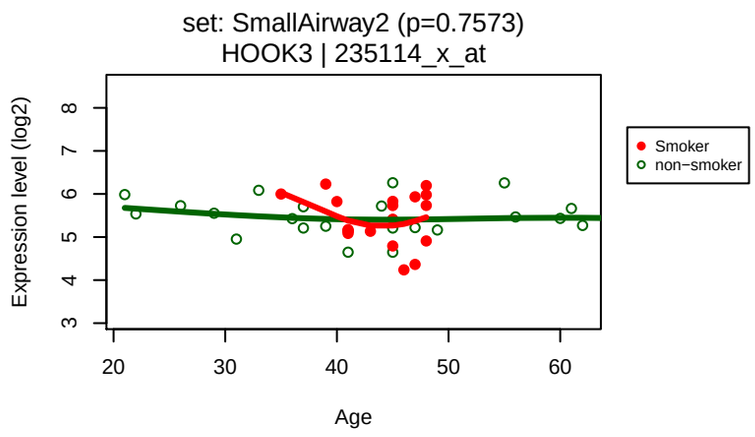
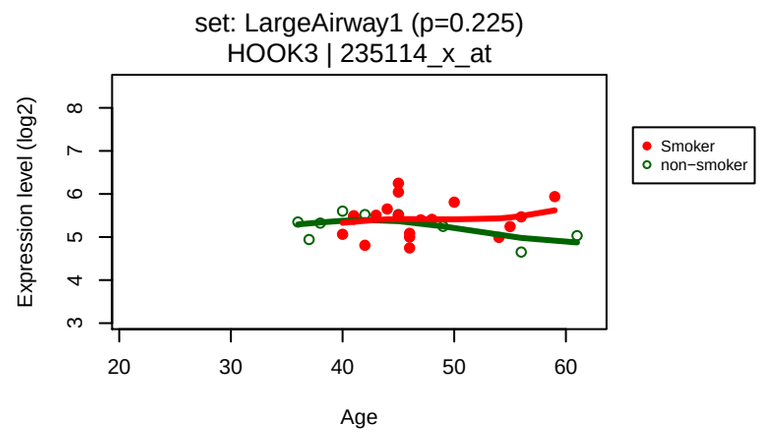

Gene expression of smokers and nonsmokers
gene: HOOK3 | 235114_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3659
p=0.7573
p=0.7502
p=0.225
p=0.5796
p=0
p=0.6844
p=0.0932
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

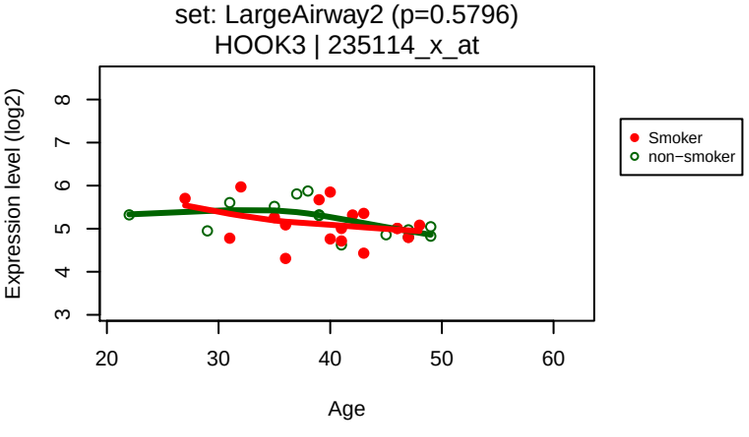

Gene expression ratio (smokers/nonsmokers)
gene: HOOK3 | 235114_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3659
p=0.7573
p=0.7502
p=0.225
p=0.5796
p=0
p=0.6844
p=0.0932
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

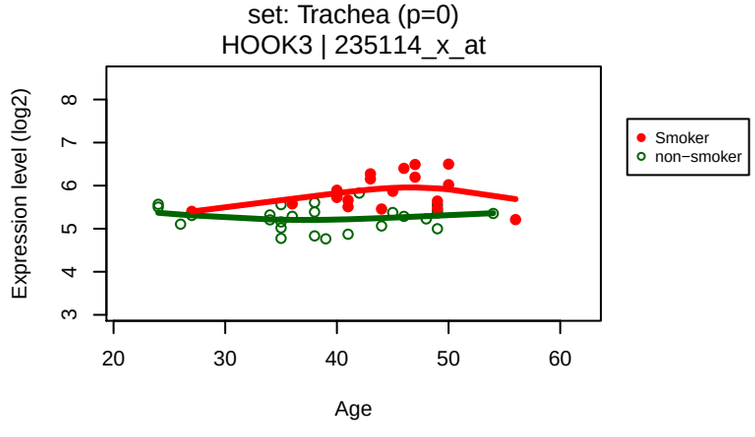

Gene expression ratio in Males
gene: HOOK3 | 235114_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8301
p=0.0265
p=1e−04
p=0.0837
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOOK3 | 235114_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.573
p=0.0268
p=0.0259
p=0.9034
SmallAirway2
LargeAirway2
Trachea
Macrophage2

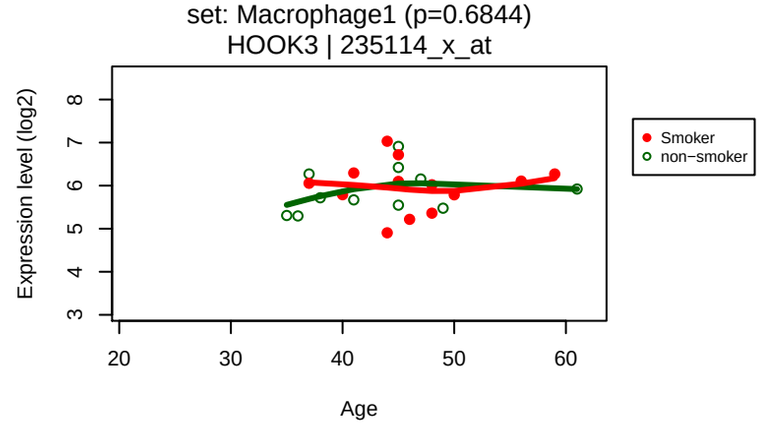

set: SmallAirway1 (p=0.3659)
HOOK3 | 235114_x_at
Expression level (log2)
Age
Smoker
non−smoker

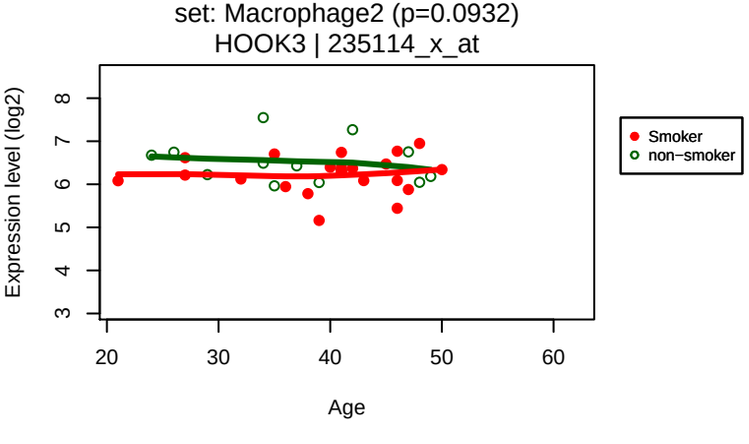

set: SmallAirway2 (p=0.7573)
HOOK3 | 235114_x_at
Expression level (log2)
Age
Smoker
non−smoker

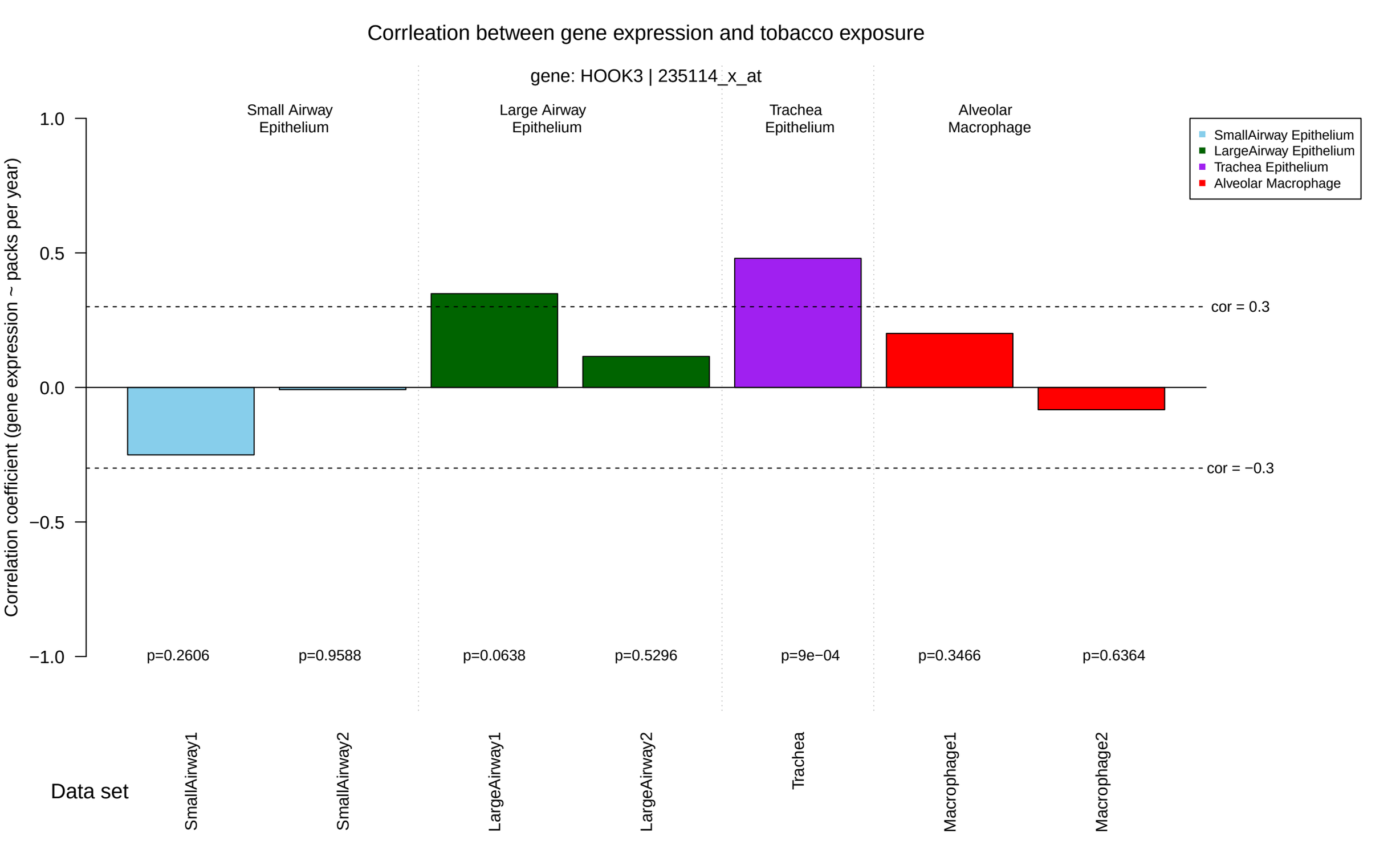

set: LargeAirway1 (p=0.225)
HOOK3 | 235114_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5796)
HOOK3 | 235114_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
HOOK3 | 235114_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6844)
HOOK3 | 235114_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0932)
HOOK3 | 235114_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOOK3 | 235114_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2606
p=0.9588
p=0.0638
p=0.5296
p=9e−04
p=0.3466
p=0.6364
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2